

Lima, de del 20..

N° Lista

NOTA

En números

En letras

Nombre del Profesor

Firma del Profesor

Solución (01)

$$\Sigma F = m a$$

$$F_B - mg = i L B - mg$$

$$a = \frac{i L B - mg}{m}$$

$$a = \frac{(2)(1,50)(1,60) - (3)}{(3 / 9,80)}$$

$$a = 5,88 \text{ m/s}^2$$

Solución (02)

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

$$\mu = NLA$$

$$\mu = NI\pi r^2$$

$$\mu = (20)(3)(0,040)^2$$

$$\mu = 0,302$$

$$\mu = 0,30 \text{ A m}^2$$

$$\tau = \mu B \sin \theta$$

$$\tau = (0,302)(0,50) \sin 60^\circ$$

$$\tau = 0,13 \text{ N.m}$$

Solución (03)

$$\vec{B} = \vec{B}_1 + \vec{B}_2$$

$$\vec{B}_1 = - \frac{\mu_0 I}{2R} \hat{i}$$

$$\vec{B}_2 = \frac{\mu_0 I}{2\pi r} \hat{i}$$

$$\vec{B} = - \frac{\mu_0 I}{2R} \hat{i} + \frac{\mu_0 I}{2\pi r} \hat{i}$$

$$\vec{B} = \left[- \frac{\mu_0 I}{2R} + \frac{\mu_0 I}{2\pi r} \right] \hat{i}$$

$$- \frac{\mu_0 I}{2R} + \frac{\mu_0 I}{2\pi r} = 0$$

$$- \frac{1}{R} + \frac{1}{\pi r} = 0$$

$$r = \frac{10,0}{\pi}$$

$$r = 3,18 \text{ cm}$$

Solución (04)

(a) y (b)

$$\oint_C \vec{B} \cdot d\vec{\ell} = B(r)(2\pi r)$$
$$= \mu_0 I_c$$
$$= \mu_0 I(r)$$

$$B(r) = \frac{\mu_0 I(r)}{2\pi r}$$

$$B(r) = \frac{\mu_0 (50) r}{2\pi r}$$

$$B(r) = \frac{\mu_0 (50)}{2\pi}$$

$$B(5,0 \text{ cm}) = B(10 \text{ cm})$$
$$= \frac{(4\pi \times 10^{-7})(50)}{2\pi}$$

$$B(5,0 \text{ cm}) = 10 \mu \text{ T}$$

$$\textcircled{c} \oint_C \vec{B} \cdot d\vec{l} = B(r) (2\pi r)$$

$$= \mu_0 I_c$$

$$= \mu_0 I (R)$$

$$B(r) = \frac{\mu_0 I (R)}{2\pi r}$$

$$B(20\text{cm}) = \frac{(4\pi \times 10^{-7}) (50) (0,10)}{2\pi (0,20)}$$

$$\underline{B(20\text{cm}) = 5,0 \mu\text{T}}$$